National University of Computer and Emerging Sciences
Lahore Campus

Programming Fundamentals

Deadline: December 1, 2025

Course Instructor
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Assignment 4

Total Time (Days):	**7**
Total Marks:	**70**
Total Questions:	**5**

Instructions to be followed

Include comments in your code wherever necessary to improve clarity. The use of Artificial Intelligence tools or Large Language Models (LLMs), such as ChatGPT or DeepSeek, is strictly forbidden. Any student found using AI or LLMs to complete this assignment will receive **zero marks** and may face **serious disciplinary action (DC)** in the future. It is better to submit some work (your effort) than submitting no solution or someone's solution.

If you are unsure about any part of the assignment, consult your **TA for clarification** before making assumptions.

Submission Instructions

Solve each question in a **separate C++ file**. Name each file using this format (RollNumber_Q1.cpp, RollNumber_Q2.cpp) Example: 25L_XXXX_Q1.cpp, 25_XXXX_Q2.cpp and so on. Put all your C++ files into **one ZIP file** with the following format **(25L_XXXX.zip)** and submit that **ZIP File** on Google Classroom.

Important: *Submissions that do not follow these instructions will result in* ***deduction of marks****. It is your duty to submit working code on GCR following the format given above. Students are expected to ensure that their programs* ***handle edge cases*** *with a user-friendly output if needed. Also handle* ***unwanted values & errors****.*

The syllabus includes character arrays, 2D Arrays and topics covered before this. The use of advanced topics will lead to deduction of marks. You can not use std::string or any form of non-primitive data types in the assignment.

Your code must be highly ***modular*** *in nature. Code snippets that can be reused and seem complex to read & understand must be encapsulated in a function to increase readability and reusability.*

Flip the Page for Questions

Question 01

You are given 2 Matrices. Your Task is to do the following operations on the matrix. *Matrices can be Rectangular or Column Matrix. Plus, there must be a function for each task.*

1. **Display** the Matrices Side by Side.
2. Calculate the **Product** of the Matrices and **Display** the Resultant Matrix
3. Calculate the **Addition** & **Difference** of the Matrices individually and display the resultant matrix.
4. Calculate the **Determinant** of a Matrix
5. **Sort** the Matrix.

Question 02

You're supposed to take 2 inputs. ***text*** and ***pattern***. Your task is to return the matching **indices** of pattern in text.

Your pattern can have the following wildcards

?	Matches a single character (one)
*	Matches all characters (zero, one or many)

Example 1
text = "wild world is when we are wrong"
pattern = "w*d"
answer = 0, 5
Explanation: *w*d means the phrase should start with w, can anything between it and should end with d.*

Example 2
text = "wild world is when we are wrong"
pattern = "w??d"
answer = 0
Explanation: *w??d means the phrase should start with w, have exactly 2 characters between it and should end with d.*

Question 03

You are given 2 very large corpuses of files. **Dataset.txt** & **needle.txt**. Your task is to return the frequency of the content in **needle.txt** in the **Dataset.txt**. No Need to do File Handling, you can simply copy paste input from Text File into the Console. And Run your Algorithm against it.

Question 04

Let's play a Game, A Classic game of Encryption. You are given a C-String, ***C** cannot exceed 5000 characters*. Ask the user for encryption key, **K** (shall be ascii). Now perform the following transformation on the content on **C**.

1. Add $\mathbf{C_i}$ Character with $\mathbf{K_i}$ (Encryption Key) supplied by the user.
2. If the $\mathbf{K_i}$ reaches the end of encryption key, Start again from the start (index=0) ($\mathbf{K_0}$) of encryption key.
3. Make sure the resultant ASCII value $\mathbf{C_i}$ remains within displayable characters after the transformation.

After doing this transformation to each character of **C**, Write the Transformed ASCII Characters back to screen. Now it's time for decryption. Your task is to reverse the encryption process by doing the transformation process in opposite and write the decrypted text on the screen again.

Question 05

You are given a string. Your task is to compress the string using the following transformation. Each character of the string shall be followed by the consecutive frequency e.g. *aaaabbccda* should be *a4b2c2d1a1*. After compressing the string, you have to decompress the string as well and print on the screen.

I hope you all will learn something new
while doing this Assignment
Best of Luck Guys
See you all energetic than before in Evaluation
Best Wishes